

Answers learning task 15

2026 – 2027

Exercise 9.1: State whether each of the following situations is a correctly stated hypothesis testing problem and why.

(a) $H_0 : \mu = 25, H_1 : \mu \neq 25$

(b) $H_0 : \sigma > 10, H_1 : \sigma = 10$

(c) $H_0 : \bar{x} = 50, H_1 : \bar{x} \neq 50$

(d) $H_0 : p = 0.1, H_1 : p = 0.5$

(e) $H_0 : s = 20, H_1 : s < 20$

Answer

(a) Correct.

(b) Incorrect, swap H_0 and H_1 .

(c) Incorrect, $\bar{x}$ is a sample statistic, not a population parameter.

(d) H_1 is incorrect, $H_1 : p \neq 0.1$ is correct.

(e) Incorrect, s is a sample statistic, not a population parameter.

Exercise 9.5: A textile fiber manufacturer is investigating a new drapery yarn, which the company claims has a mean thread elongation of 12 kg with a standard deviation of 1.5 kg. The company wishes to test the hypothesis $H_0 : \mu = 12$ against $H_1 : \mu < 12$, using a random sample of four specimens.

(a) What is the type I error probability if the critical region is defined as $\bar{x} < 11.5$ kg?

(b) Find β for the case in which the true mean elongation is 11.25 kg.

(c) Find β for the case in which the true mean is 11.5 kg.

Answer

Note: Calculations use $\sigma = 0.5$ kg.

(a) 0.0228

(b) 0.1587

(c) 0.5000

Exercise 9.7: In Exercise 9.5, find the boundary of the critical region if the type I error probability is

(a) $\alpha = 0.01$ and $n = 5$

(c) $\alpha = 0.01$ and $n = 20$

(b) $\alpha = 0.05$ and $n = 5$

(d) $\alpha = 0.05$ and $n = 20$

Answer

(a) 11.4798

(b) 11.6322

(c) 11.7399

(d) 11.8161

Exercise 9.9: In Exercise 9.5, calculate the p -value if the observed statistic is(a) $\bar{x} = 11.25$ (b) $\bar{x} = 11.0$ (c) $\bar{x} = 11.75$ **Answer**

(a) 0.0013

(b) 0.000032

(c) 0.1587

Exercise 9.15: A consumer products company is formulating a new shampoo and is interested in foam height (in millimeters). Foam height is approximately normally distributed and has a standard deviation of 25 mm. The company wishes to test $H_0 : \mu = 175$ mm versus $H_1 : \mu > 175$ mm, using the results of $n = 15$ samples.(a) Find the type I error probability α if the critical region is $\bar{x} > 185$.

(b) What is the probability of type II error if the true mean foam height is 185 mm?

(c) Find β for the true mean of 195 mm.**Answer**

(a) 0.0607

(b) 0.5

(c) 0.0607

Exercise 9.17: In Exercise 9.15, find the boundary of the critical region if the type I error probability is(a) $\alpha = 0.01$ and $n = 10$ **Answer**

193.3914

Exercise 9.19: In Exercise 9.15, calculate the p -value if the observed statistic is(a) $\bar{x} = 180$ (b) $\bar{x} = 190$ (c) $\bar{x} = 170$ **Answer**

(a) 0.2193

(b) 0.0101

(c) 0.7807

Exercise 9.31: A hypothesis will be used to test that a population mean equals 4 against the alternative that the population mean is more than 4 with known variance σ . What is the critical value for the test statistic Z_0 for the following significance levels?

(a) 0.01

(b) 0.05

(c) 0.10

Answer

(a) 2.3263

(b) 1.6449

(c) 1.2816

Exercise 9.33: For the hypothesis test $H_0 : \mu = 6$ against $H_1 : \mu \neq 6$ and variance known, calculate the p -value for each of the following test statistics.

(a) $z_0 = 2.05$

(b) $z_0 = -1.84$

(c) $z_0 = 0.4$

Answer

(a) 0.0404

(b) 0.0658

(c) 0.6892